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Car Rental Platform

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1. Project Overview

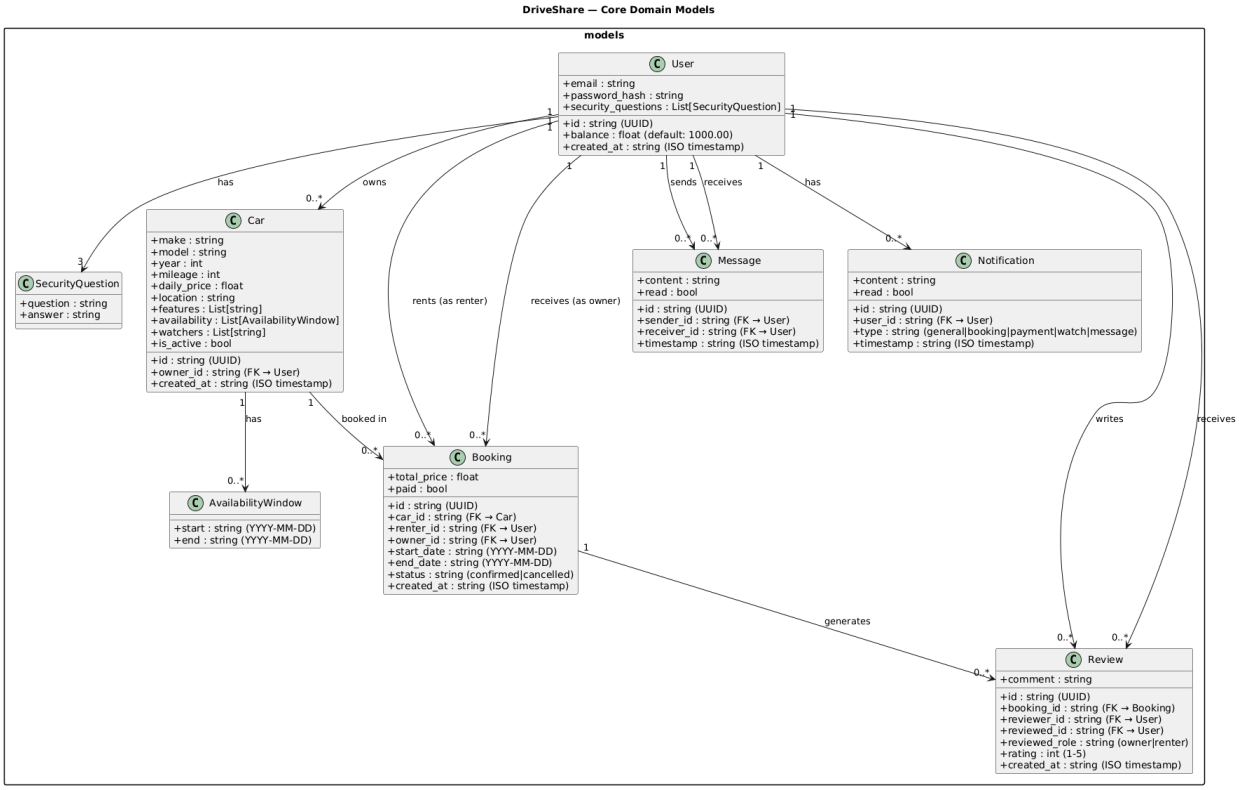
DriveShare is a peer-to-peer car rental platform inspired by Turo.com. The application allows car owners to list their vehicles for rent and renters to search, book, and pay for rentals. The system implements six design patterns and features a React frontend with a Python/FastAPI backend using SQLite for persistence.

Key Features:

- User registration and authentication with SHA-256 password hashing
- Security questions for password recovery (Chain of Responsibility pattern)
- Car listing management with Builder pattern
- Search and booking with overlap prevention
- Watch system for price drop notifications (Observer pattern)
- In-app messaging between users
- Simulated payment processing (Proxy pattern)
- Session management (Singleton pattern)
- UI component coordination (Mediator pattern)

2. Class Diagrams

2.1 Domain Models

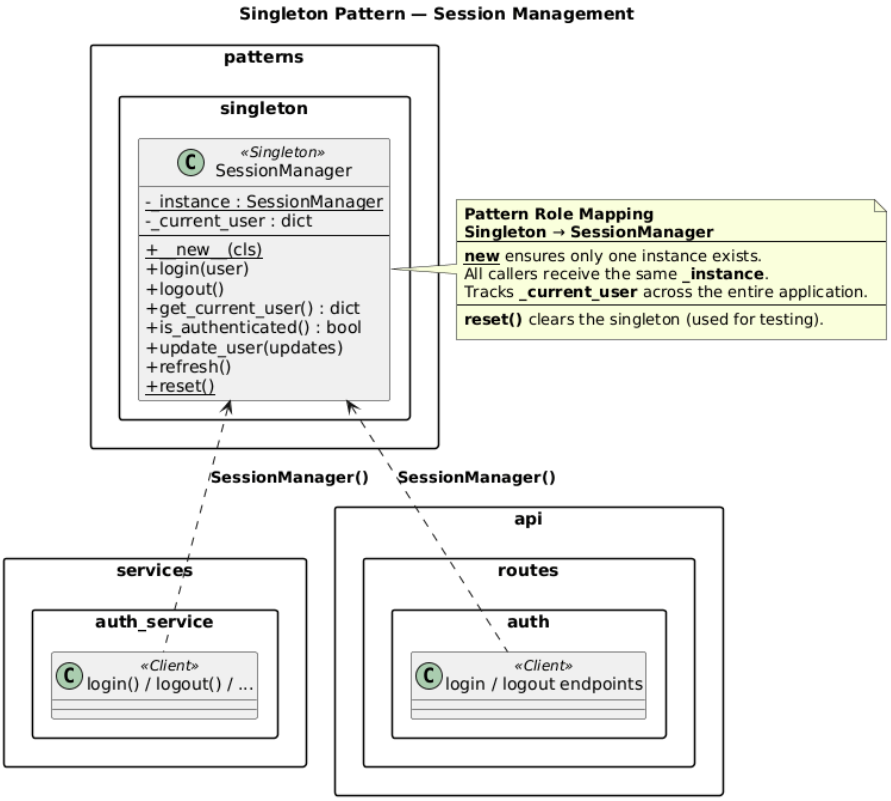


Description: The Core Domain Model diagram illustrates the main entities in the DriveShare system and their relationships:

- **User:** Central entity representing both car owners and renters. Each user has 3 security questions for password recovery and a starting balance of $1,000.
- **Car:** Represents a vehicle listing with details like make, model, year, pricing, and availability windows. Linked to its owner and can have multiple watchers (users monitoring price changes).
- **Booking:** Captures a rental transaction between a renter and owner for a specific car and date range. Tracks payment status.
- **Message:** Enables direct communication between users (typically renter and owner).
- **Notification:** System-generated alerts for bookings, payments, price drops, and messages.
- **Review:** Post-rental feedback where renters review owners and vice versa.

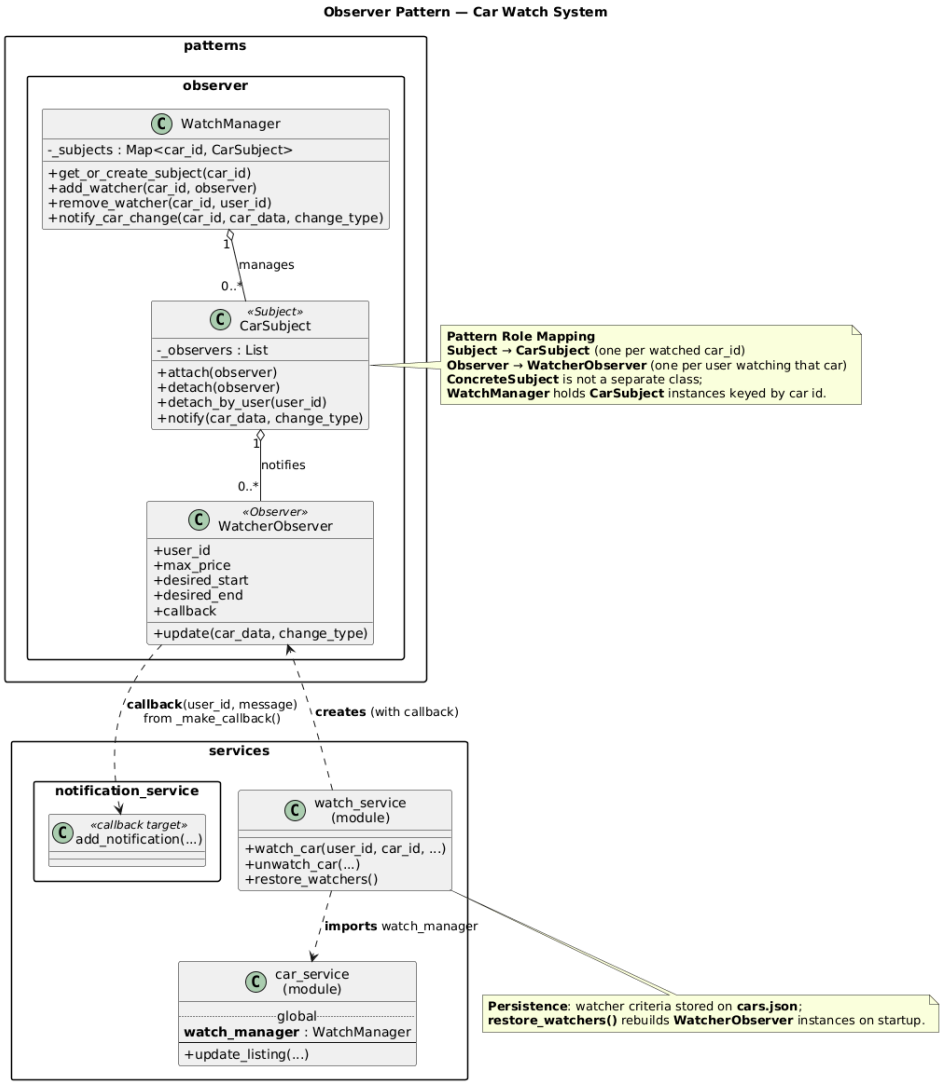
2.2 Design Patterns

2.2.1 Singleton Pattern: Session Management



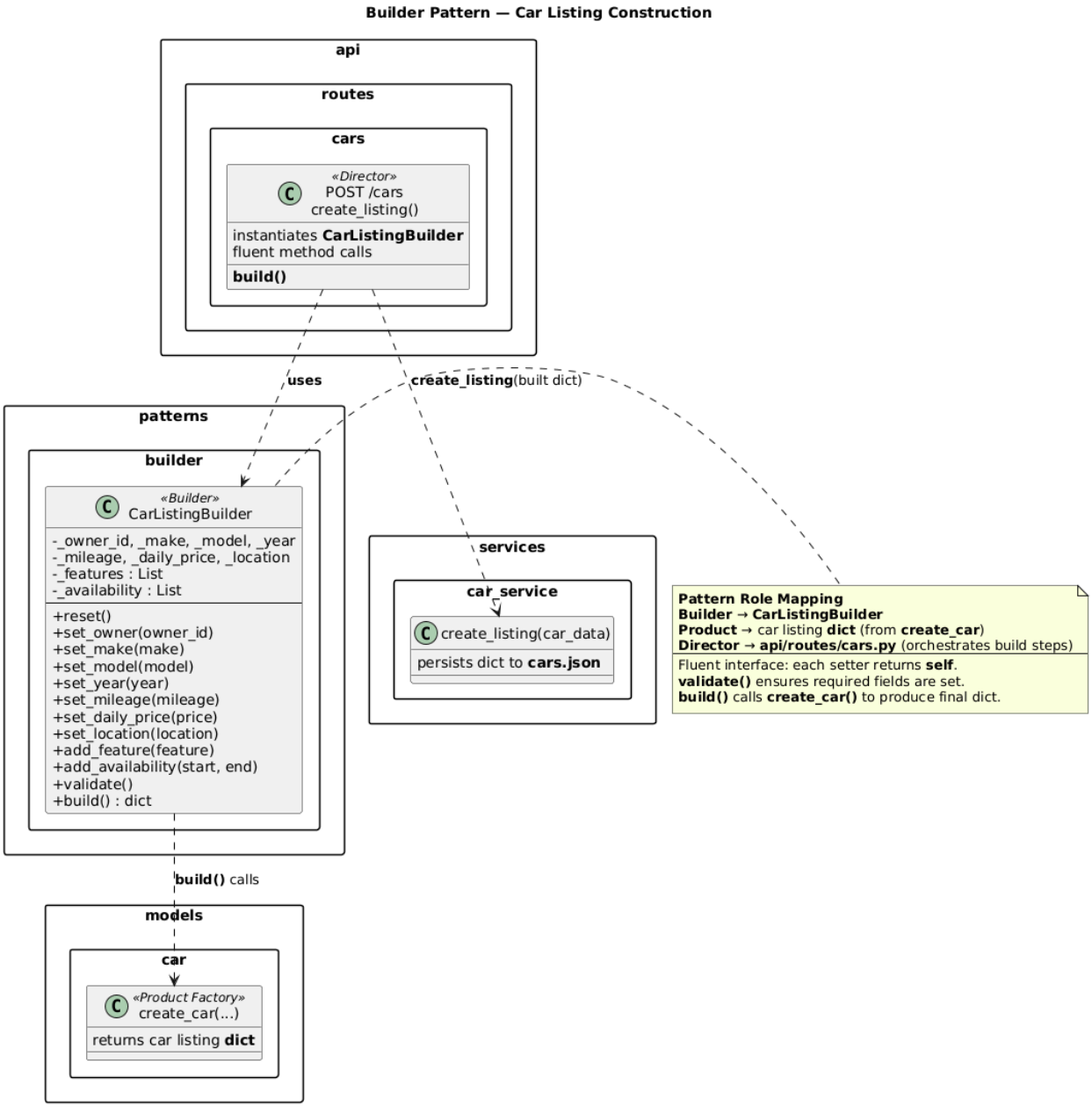
Description: The Singleton pattern ensures only one SessionManager instance exists throughout the application lifecycle. This is critical for maintaining a consistent user session state across all components. The **__new__** method intercepts object creation and returns the already existing instance if one exists. Key operations include **login()** to set the current user, **logout()** to clear the session, and **refresh()** to reload user data from storage (useful after balance changes from payments).

2.2.2 Observer Pattern: Car Watch System



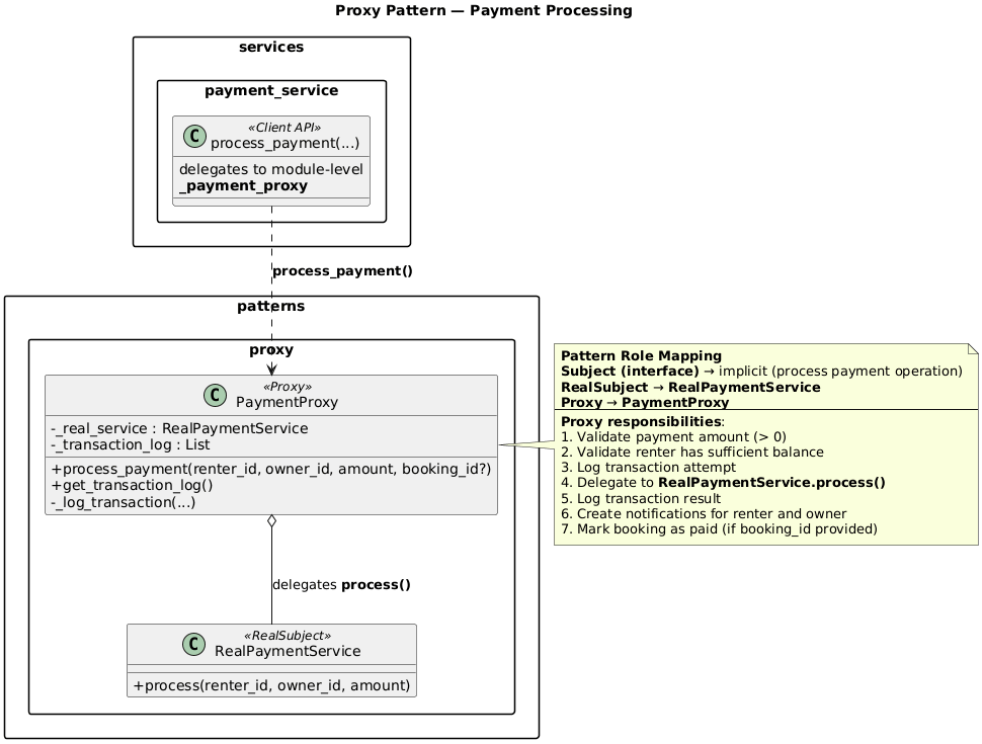
Description: The Observer pattern implements the "Watch" feature where renters can subscribe to car listings and receive notifications when prices drop or availability changes. Each CarSubject maintains a list of WatcherObserver instances. When car_service.update_listing() detects a price drop, it calls WatchManager.notify_car_change(), which triggers the update() method on all observers. Each observer evaluates the change against their criteria (e.g., `max_price`) and uses a callback to create a notification if the criteria are met.

2.2.3 Builder Pattern: Car Listing Construction



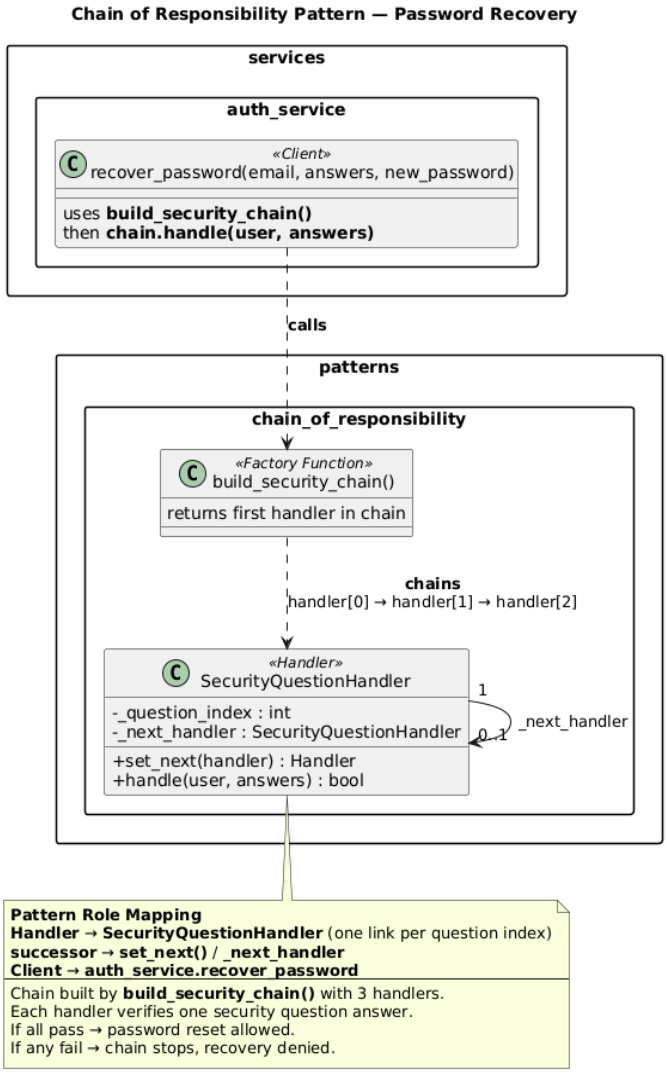
Description: The Builder pattern provides a step-by-step, fluent interface for constructing complex car listing objects. This is beneficial because car listings have many optional fields (features, availability windows) and required fields (make, model, year, price, location). The CarListingBuilder accumulates field values through chained setter calls, validates that all required fields are present via validate(), and finally produces the car dictionary via build(). The API route acts as the "Director" that orchestrates the build process based on user input.

2.2.4 Proxy Pattern: Payment Processing



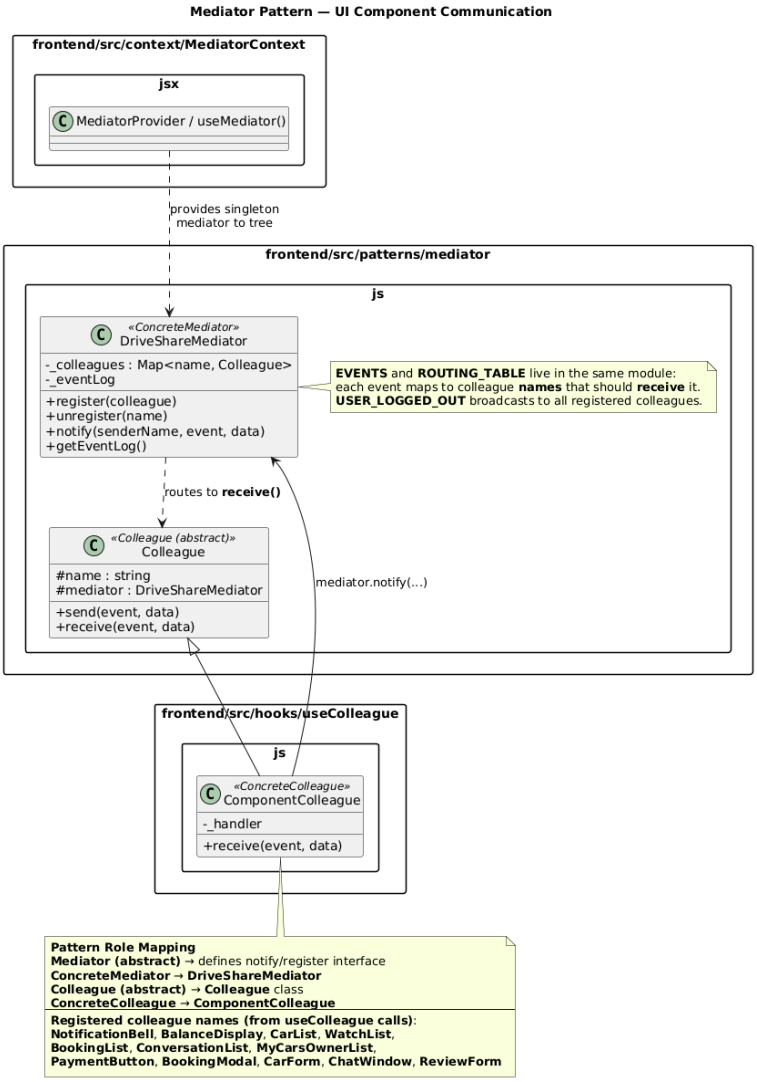
Description: The Proxy pattern adds validation, logging, and notification layers around the payment processing logic. The PaymentProxy acts as a middleman: it first validates that the payment amount is positive and that the renter has sufficient balance. If validation passes, it delegates to `RealPaymentService.process()` which performs the actual balance transfers. The proxy then logs the transaction (success or failure), creates notifications for both renter and owner, and marks the associated booking as paid. This separation of concerns keeps the core payment logic simple while adding cross-cutting concerns transparently.

2.2.5 Chain of Responsibility Pattern: Password Recovery



Description: The Chain of Responsibility pattern implements the multi-step password recovery process. Each SecurityQuestionHandler is responsible for verifying one of the user's three security question answers. The handlers are linked via set_next() to form a chain. When handle() is called, the first handler checks if the first answer is correct. If so, it passes control to the next handler; if not, it returns False immediately. This continues until all three answers are verified (returns True) or any answer fails (returns False). The factory function build_security_chain() constructs the chain with handlers for indices 0, 1, and 2.

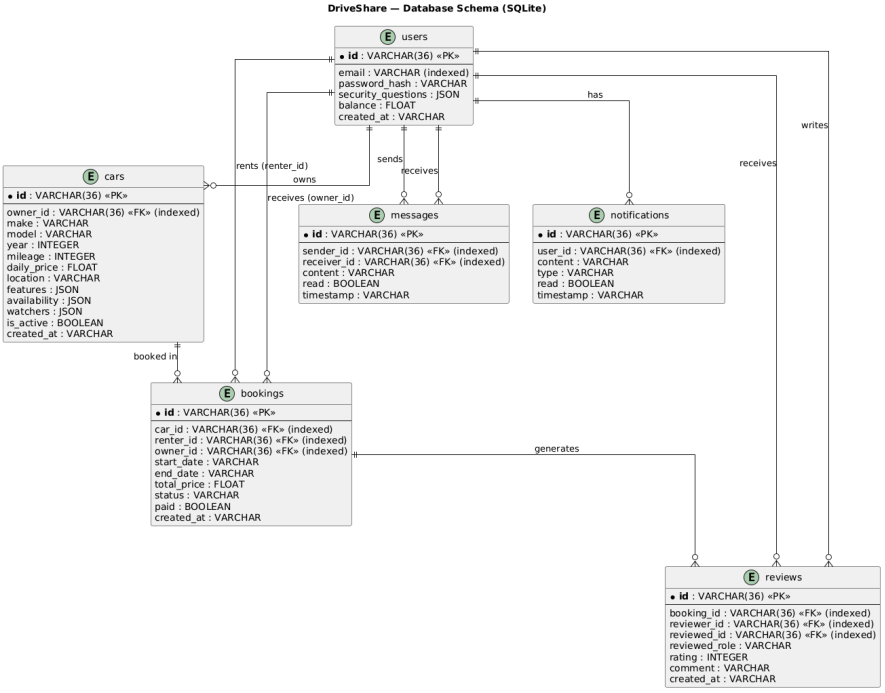
2.2.6 Mediator Pattern: UI Component Communication



Description: The Mediator pattern decouples React UI components by centralizing their communication through DriveShareMediator. Instead of components directly calling each other, they register as "colleagues" with the mediator and communicate via events. For example, when PaymentButton processes a payment, it sends a BOOKING_PAID event. The mediator routes this to BalanceDisplay (to refresh the balance), BookingList (to update payment status), and NotificationBell (to show a new notification). This pattern eliminates tight coupling between components and makes the event flow explicit and traceable via getEventLog().

3. Database Schema

3.1 Entity-Relationship Diagram (PlantUML)



3.2 Table Descriptions

Users

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
email	VARCHAR	User's email address (indexed for login lookup)
password_hash	VARCHAR	SHA-256 hash of the password
security_questions	JSON	Array of 3 objects with question and answer fields
balance	FLOAT	User's wallet balance (starts at $1000)
created_at	VARCHAR	ISO timestamp of account creation

Cars

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
owner_id	VARCHAR(36)	Foreign key to users table
make	VARCHAR	Manufacturer (e.g., "Toyota")
model	VARCHAR	Model name (e.g., "Camry")
year	INTEGER	Model year
mileage	INTEGER	Current odometer reading
daily_price	FLOAT	Rental price per day
location	VARCHAR	Pick-up location
features	JSON	Array of feature strings (e.g., ["Bluetooth", "GPS"])
availability	JSON	Array of `{start, end}` date windows

watchers	JSON	Array of user IDs watching this car
is_active	BOOLEAN	Soft-delete flag (false = deleted)
created_at	VARCHAR	ISO timestamp of listing creation

Bookings

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
car_id	VARCHAR(36)	Foreign key to cars table
renter_id	VARCHAR(36)	Foreign key to users (person renting)
owner_id	VARCHAR(36)	Foreign key to users (car owner)
start_date	VARCHAR	Rental start date (YYYY-MM-DD)
end_date	VARCHAR	Rental end date (YYYY-MM-DD)
total_price	FLOAT	Calculated total (daily_price × days)
status	VARCHAR	"confirmed" or "cancelled"
paid	BOOLEAN	Payment completion flag
created_at	VARCHAR	ISO timestamp of booking creation

Messages

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
sender_id	VARCHAR(36)	Foreign key to users (message author)
receiver_id	VARCHAR(36)	Foreign key to users (recipient)
content	VARCHAR	Message text content
read	BOOLEAN	Read receipt flag
timestamp	VARCHAR	ISO timestamp of when message was sent

Notifications

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
user_id	VARCHAR(36)	Foreign key to users (notification recipient)
content	VARCHAR	Notification message text
type	VARCHAR	Category: "general", "booking", "payment", "watch", or "message"
read	BOOLEAN	Dismissal flag
timestamp	VARCHAR	ISO timestamp of account creation

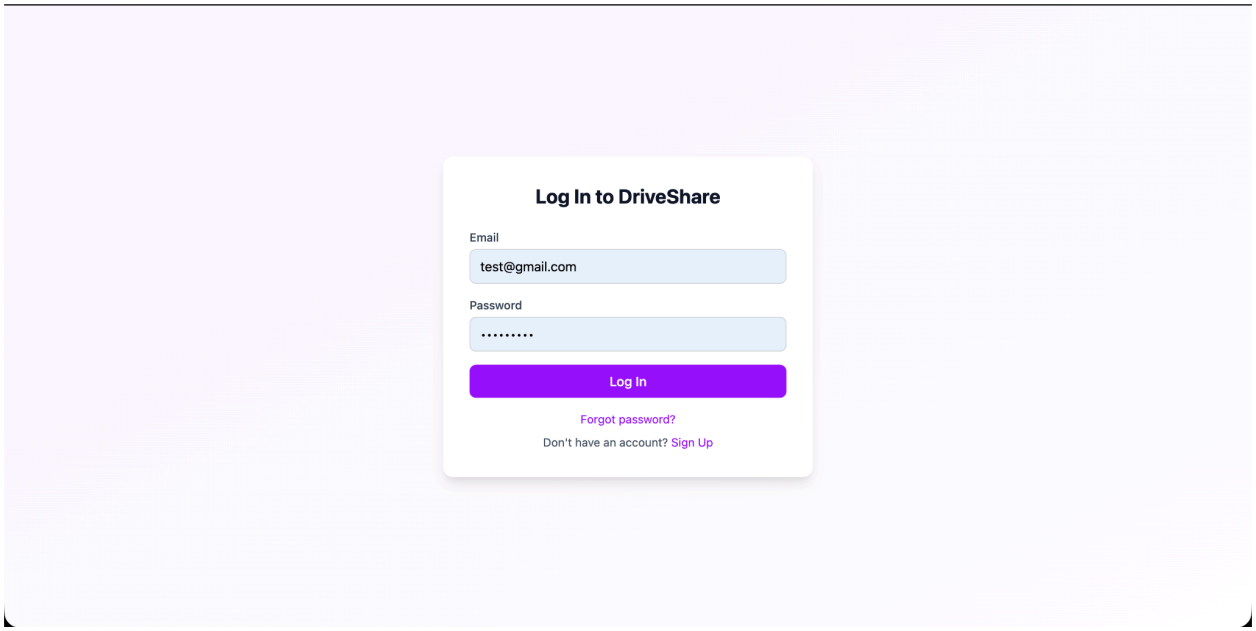
Reviews

Column	Type	Description
id	VARCHAR(36)	Primary key (UUID)
booking_id	VARCHAR(36)	Foreign key to bookings table
reviewer_id	VARCHAR(36)	Foreign key to users (person writing review)
reviewed_id	VARCHAR(36)	Foreign key to users (person being reviewed)
reviewed_role	VARCHAR	"owner" or "renter"
rating	INTEGER	Score from 1-5
comment	VARCHAR	Review text
created_at	VARCHAR	ISO timestamp of review submission

4. User Interface Screenshots

4.1 Authentication Pages

Login Page

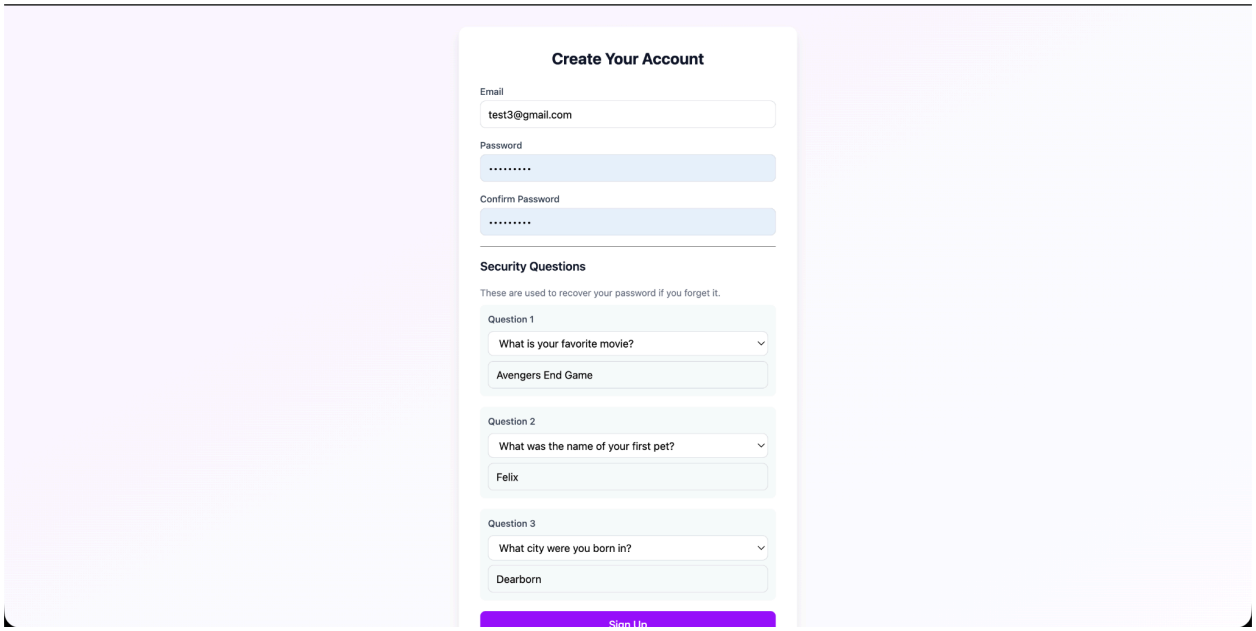


Description: A centered login form with a clean, modern design featuring a purple gradient background. The form contains:

- "Log In to DriveShare" header
- Email input field with label
- Password input field with label
- Purple "Log In" button
- "Forgot password?" link below the button
- "Don't have an account? Sign Up" link at the bottom

The form includes an error alert display for invalid credentials and loading state on the submit button.

Registration Page

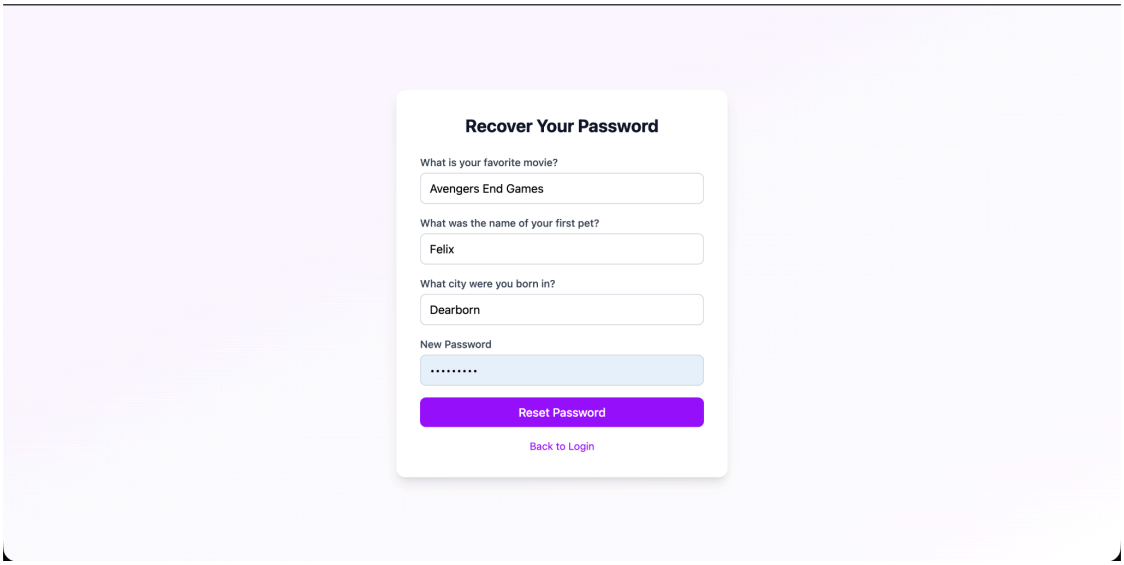


Description: A multi-step registration form with the same purple gradient styling. The form collects:

- Email address
- Password (minimum 4 characters)
- Three security questions with answer fields (for password recovery)
- "Create Account" button

The page displays validation errors inline and redirects to the dashboard upon successful registration.

Forgot Password Page



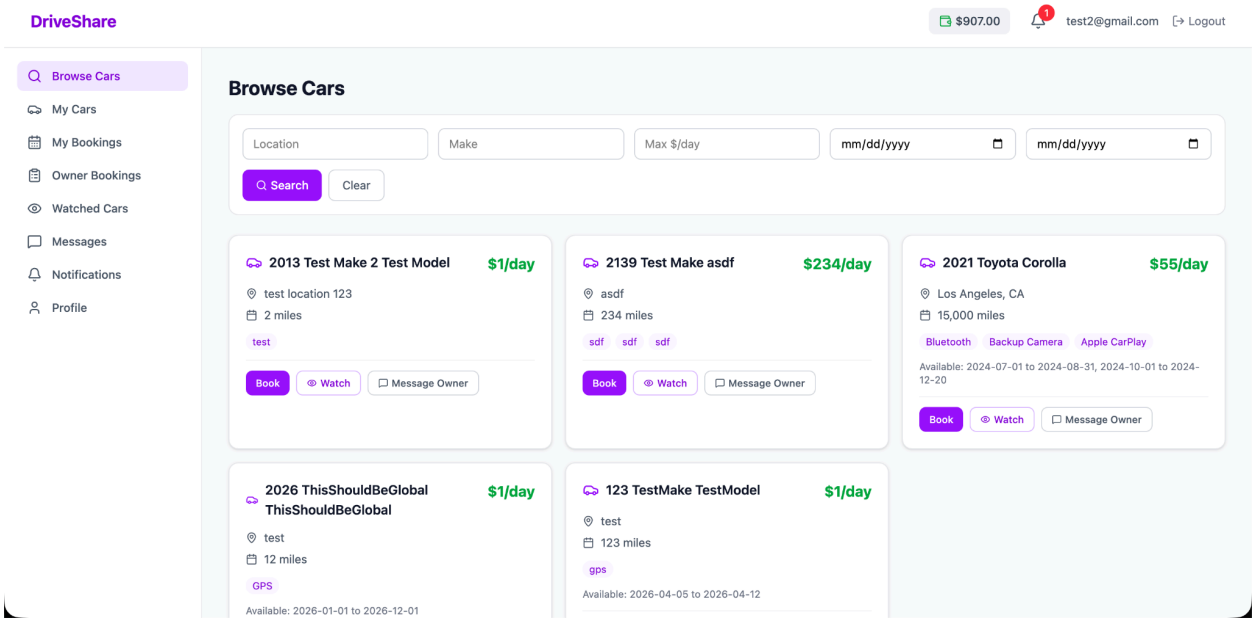
Description: A two-step password recovery interface:

- **Step 1:** User enters their email address to retrieve their security questions
- **Step 2:** User answers all three security questions and enters a new password

The Chain of Responsibility pattern is used behind the scenes to verify each answer sequentially.

4.2 Dashboard & Navigation

Dashboard Layout



Description: The main authenticated interface with:

- Top Navbar: DriveShare logo, notification bell icon (with unread count badge), user balance display ($X.XX), and logout button
- Left Sidebar: Navigation links including:
    - Browse Cars
    - My Cars
    - My Bookings
    - Owner Bookings
    - Watched Cars
    - Messages
    - Profile
- Main Content Area: Renders the selected page content via React Router outlet

The sidebar collapses on mobile with a hamburger menu toggle.

4.3 Car Management

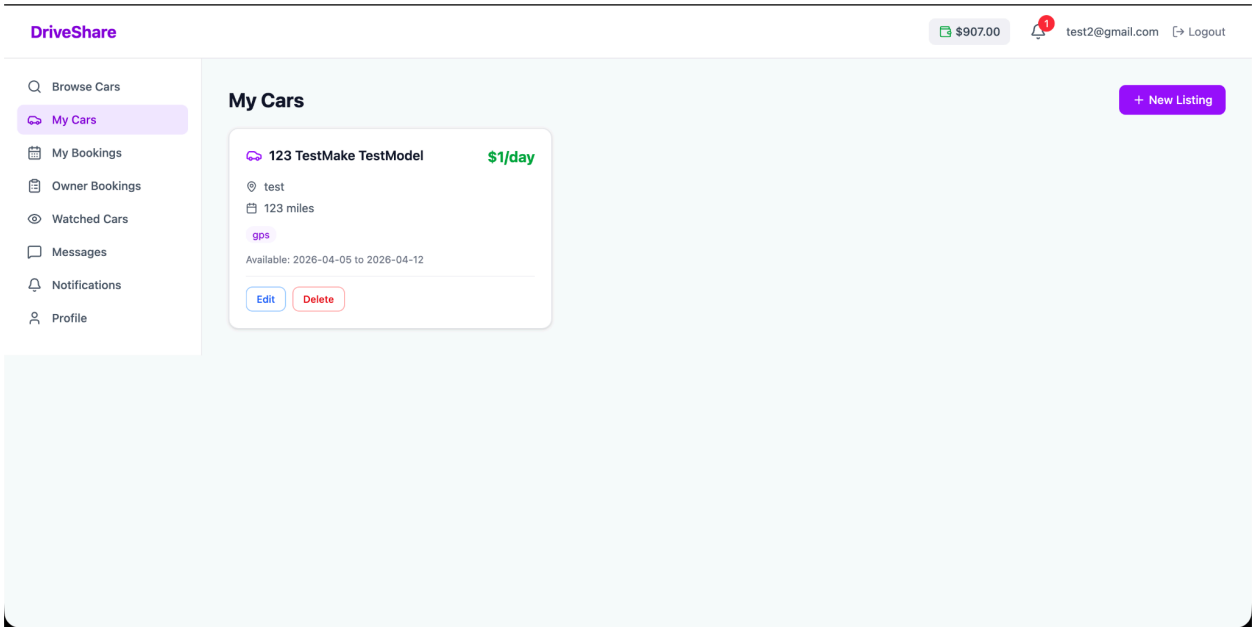
Browse Cars Page

Description: A searchable grid of available car listings featuring:

- Search Form: Filters for location, date range, max price, and make
- **Car Cards Grid:** Responsive 1/2/3 column layout showing each car with:
    - Make, model, and year
    - Daily price prominently displayed
    - Location
    - Feature tags (Bluetooth, GPS, etc.)
    - "Book" button to open booking modal
    - "Watch" button to monitor price drops
    - "Message Owner" button

The page uses the Mediator pattern to refresh when cars are added, updated, or deleted.

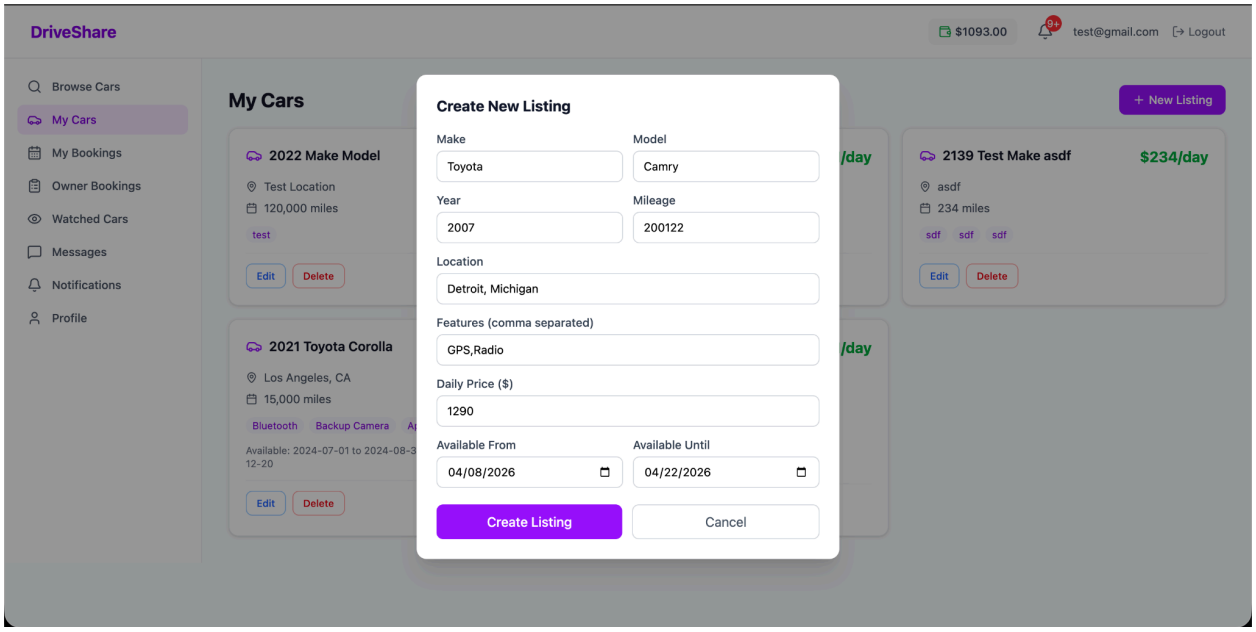
My Cars Page (Owner View)



Description: Owner's dashboard for managing their car listings with:

- "My Cars" header with "New Listing" button (purple, with plus icon)
- Grid of owned car cards, each showing:
    - Car details (make, model, year, price, location)
    - "Edit" button to modify listing
    - "Delete" button to remove listing
- Empty state message: "You haven't listed any cars yet."

Create/Edit Car Listing Form



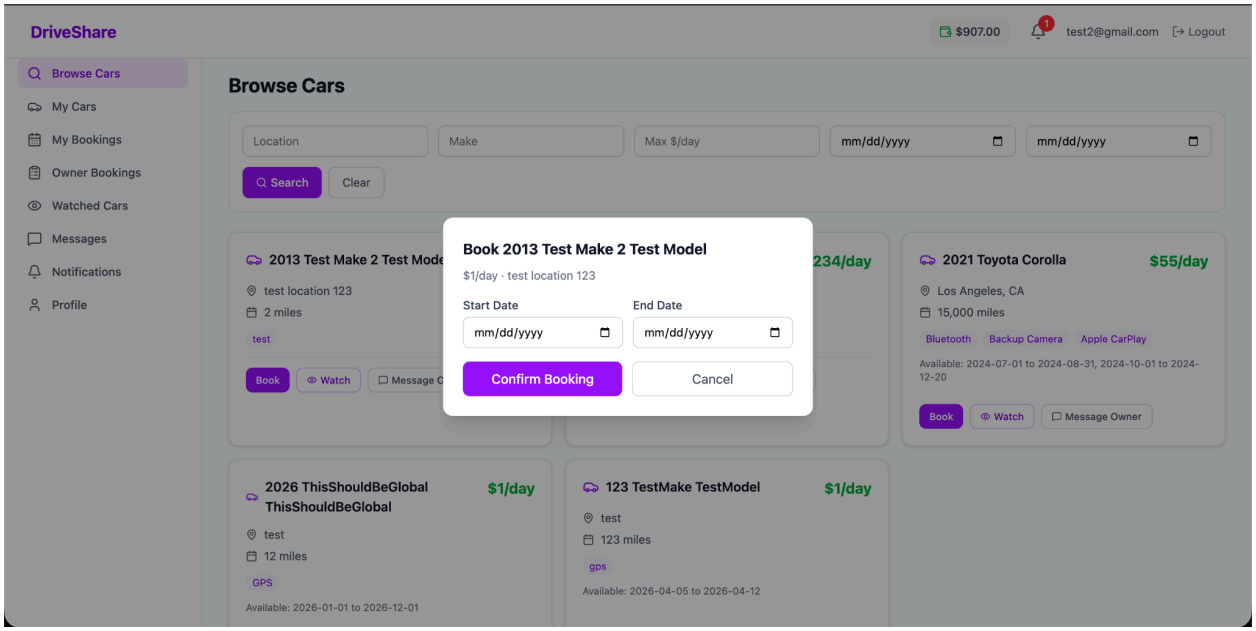
Description: A modal form for creating or editing car listings with:

- Make input field
- Model input field
- Year input field
- Mileage input field
- Daily price input field
- Location input field
- Features input (comma-separated)
- Availability date ranges (add/remove windows)
- "Save" and "Cancel" buttons

The Builder pattern is used on the backend to construct the car object from form data.

4.4 Booking System

Book a Car Modal

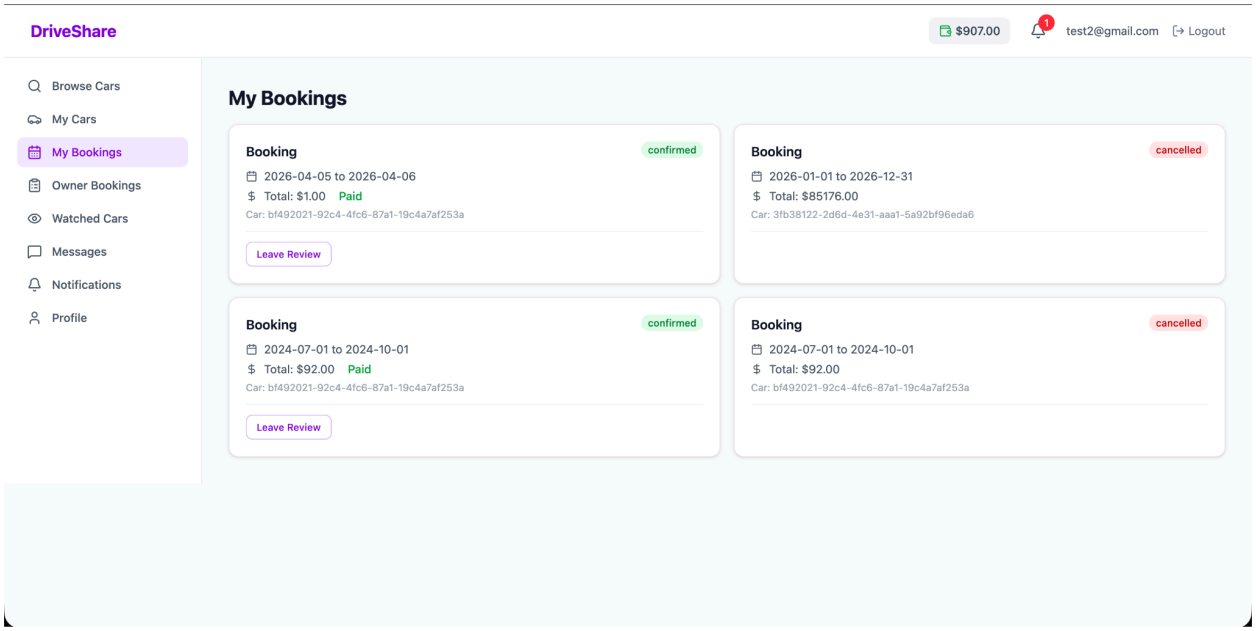


Description: A modal dialog for creating a new booking:

- Car summary at top (make, model, daily price)
- Start date picker
- End date picker
- Calculated total price display (days × daily_price)
- "Confirm Booking" button
- "Cancel" button

Includes validation for date overlap and prevents booking own cars.

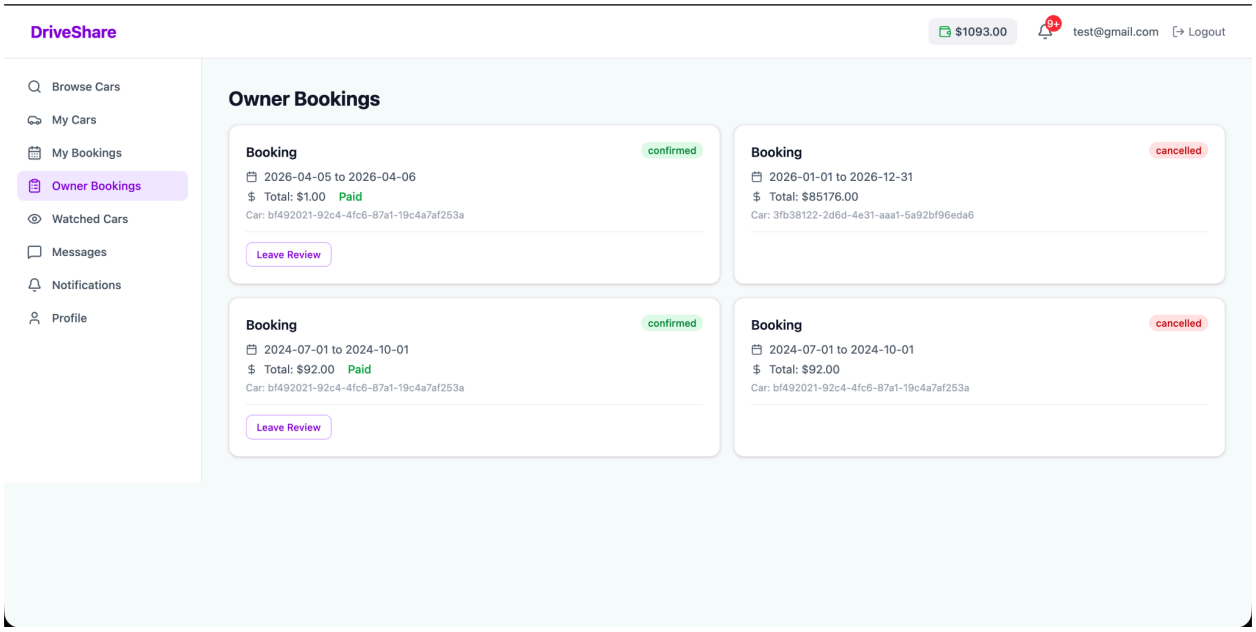
My Bookings Page (Renter View)



Description: List of bookings where the user is the renter:

- Booking cards showing:
- Car info (make, model)
- Date range (start – end)
- Total price
- Status badge (Confirmed/Cancelled)
- Payment status (Paid/Unpaid)
- "Pay Now" button (if unpaid) (uses Proxy pattern)
- "Cancel Booking" button
- "Leave Review" button (after rental complete)

Owner Bookings Page

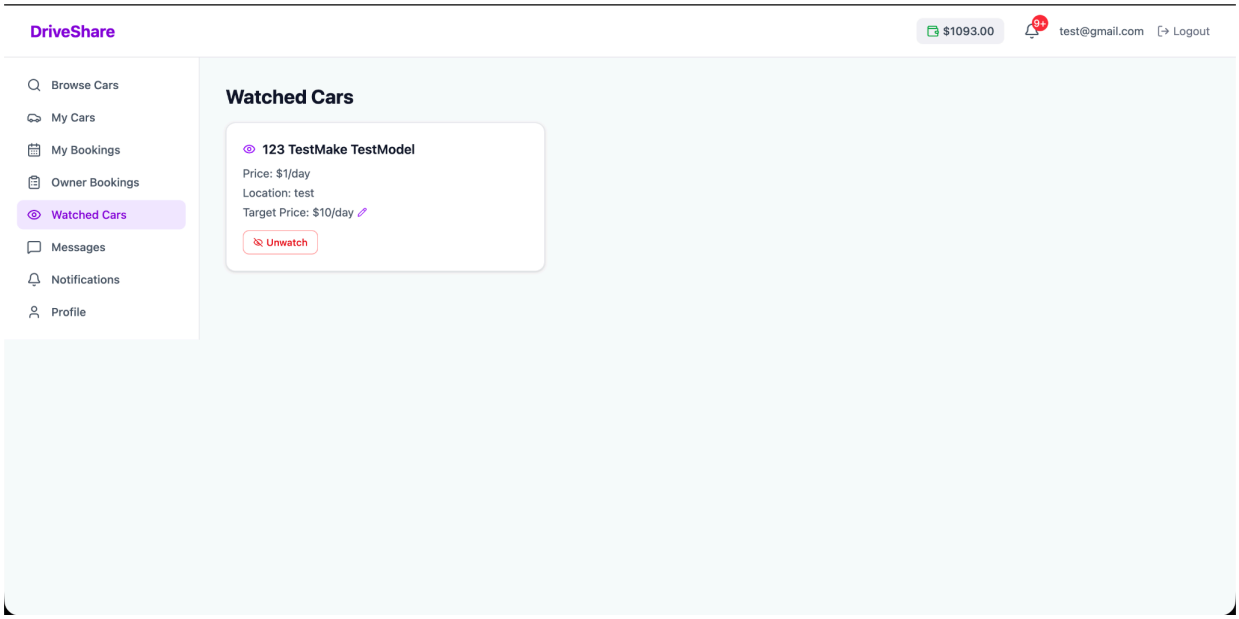


Description: List of bookings for cars the user owns:

- Similar card layout to My Bookings
- Shows renter's email
- "Message Renter" button for communication
- Payment received status

4.5 Watch & Notifications

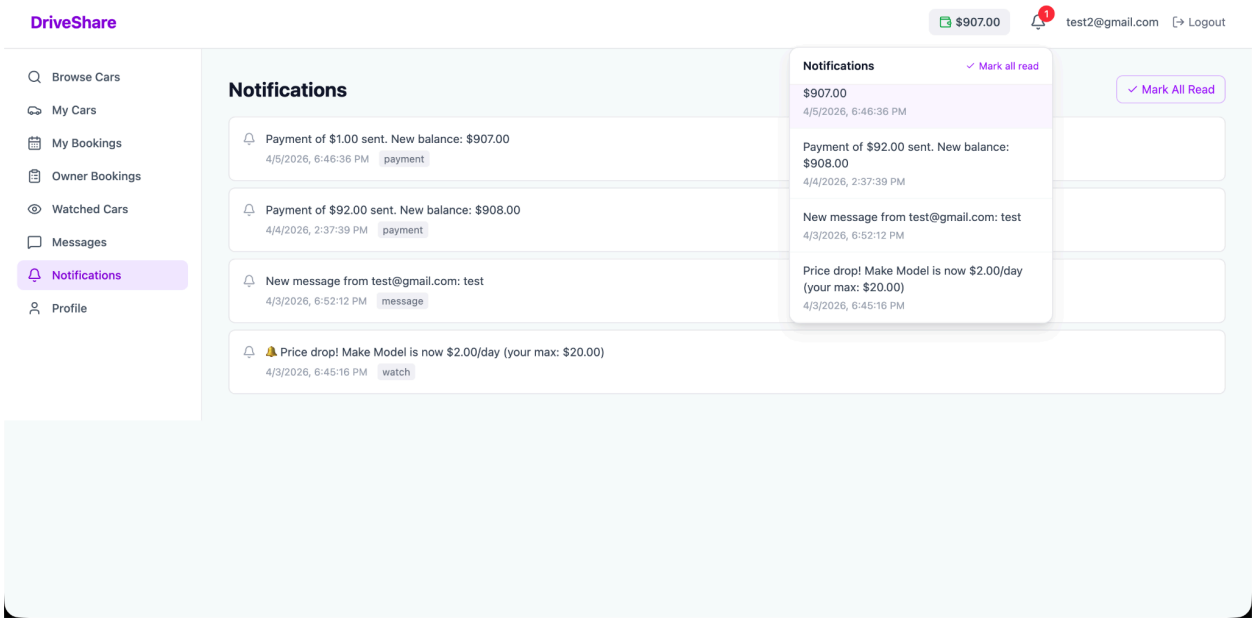
Watched Cars Page



Description: List of cars the user is watching for price changes:

- Car cards with current price
- Target price set by user
- "Unwatch" button to remove from watch list
- Notification appears when price drops below target (Observer pattern)

Notifications Panel



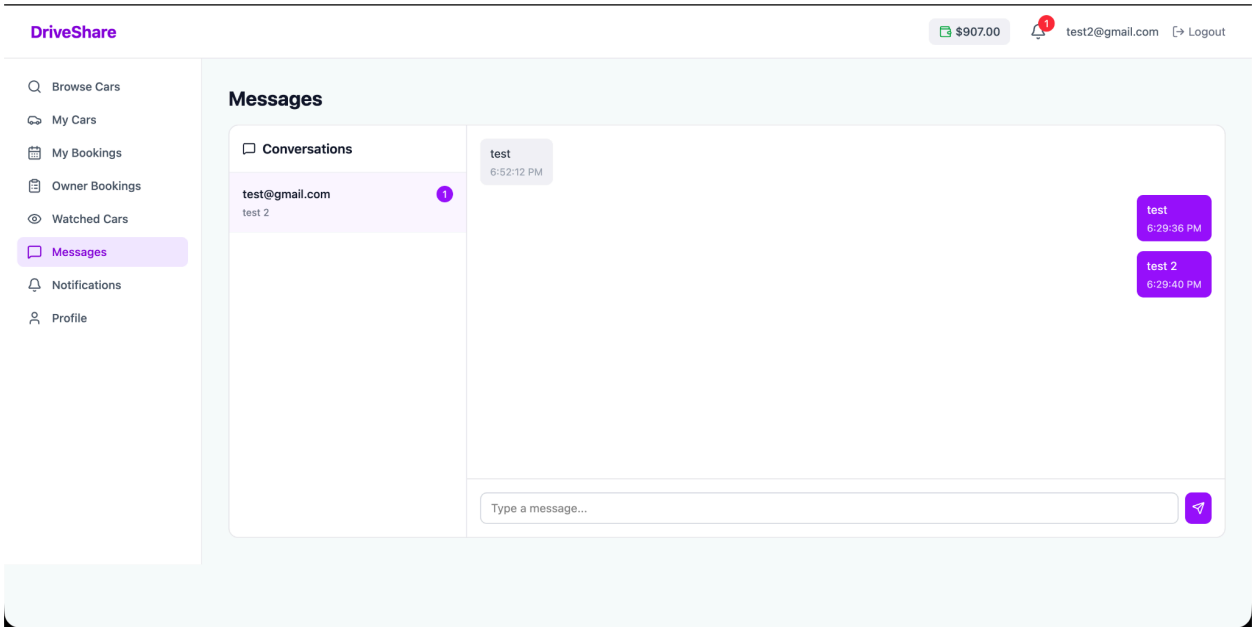
Description: Dropdown panel from the notification bell showing:

List of notifications with icons by type:

- Booking notifications
    - Payment notifications
    - Watch/price drop notifications
    - Message notifications
- Timestamp for each notification
    - "Mark as Read" action
    - Unread count badge on bell icon

4.6 Messaging

Messages Page

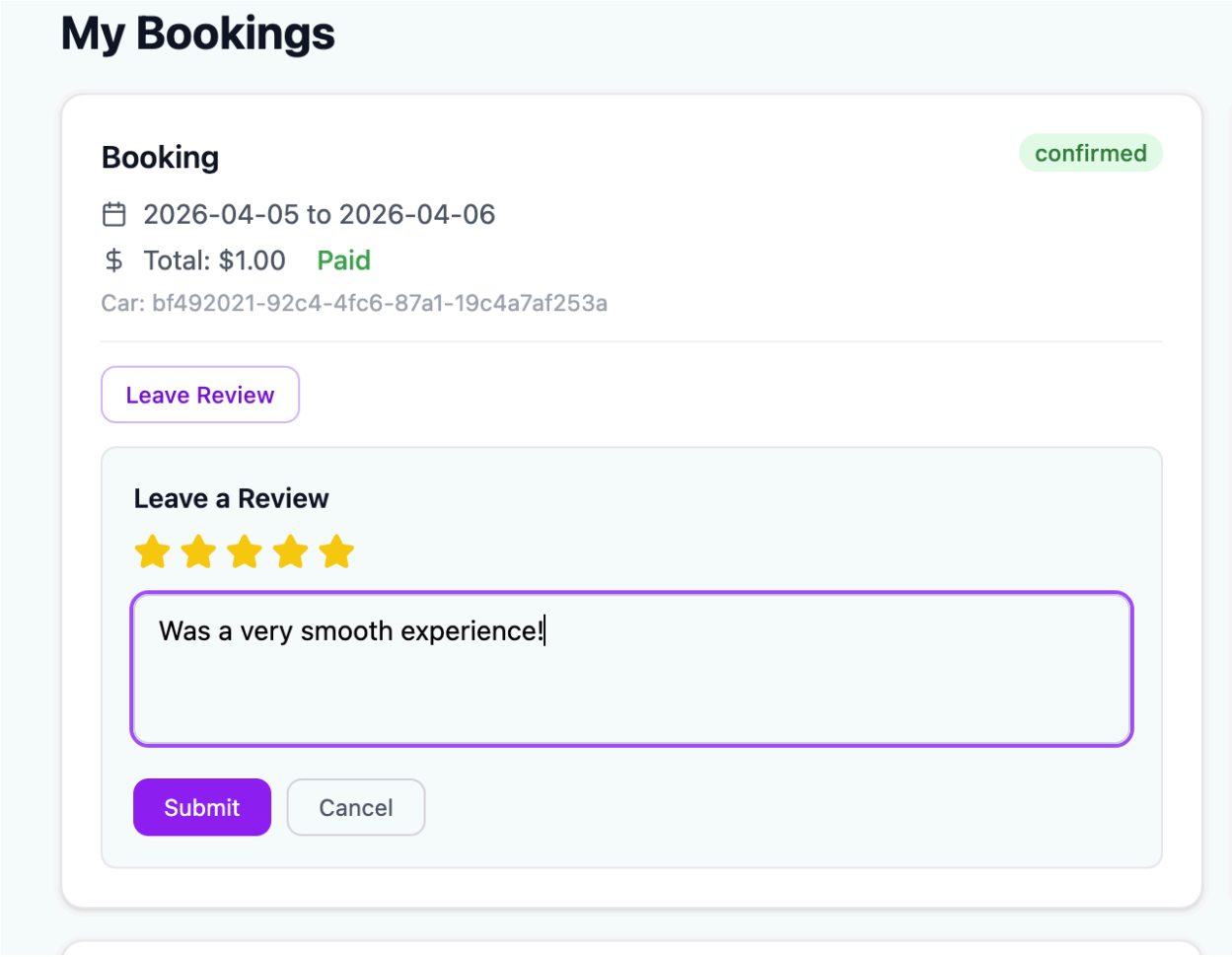


Description: Split-pane messaging interface:

- Left Panel: Conversation list showing other users with unread message count
- Right Panel: Chat window displaying: Message bubbles (sent messages on right, received on left), Timestamps, Message input field at bottom, and "Send" button

4.7 Reviews & Profile

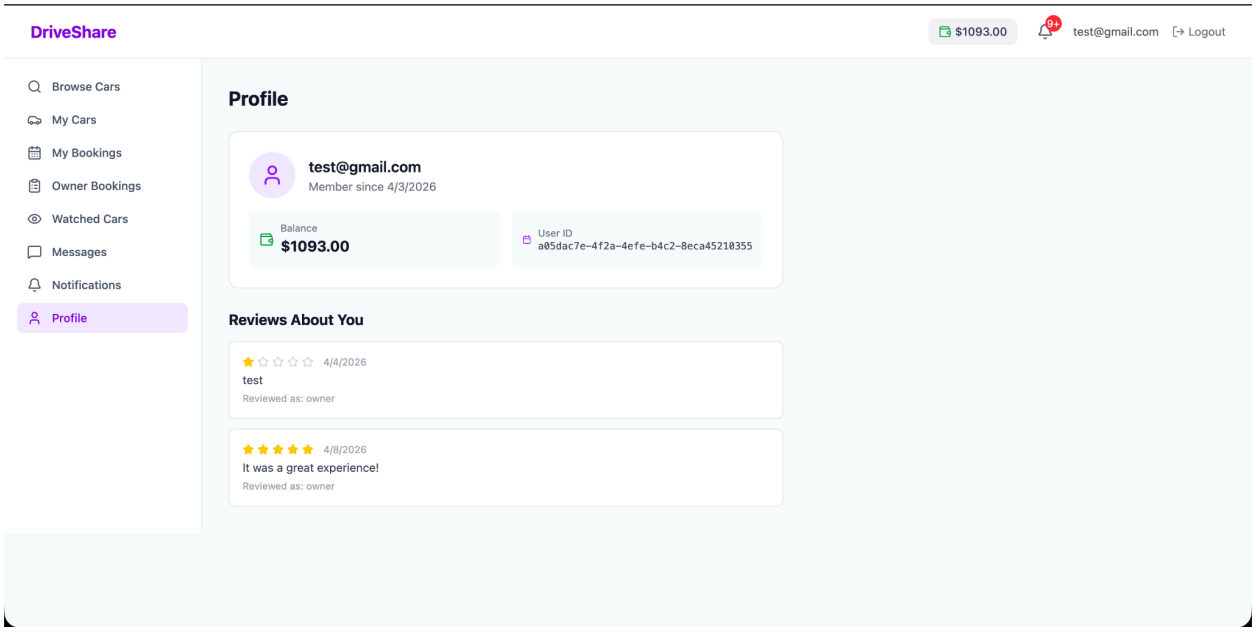
Review Form



Description: Post-rental review submission form:

- Star rating selector (1-5 stars)
- Comment text area
- "Submit Review" button
- Appears after booking is complete and paid

Profile Page



Description: User profile page showing:

- Email address
- Current wallet balance
- Reviews received (as owner and as renter)
- Average rating display

5. References

5.1 Design Patterns

1. **Refactoring Guru - Design Patterns**
    - https://refactoring.guru/design-patterns/singleton
    - https://refactoring.guru/design-patterns/observer
    - https://refactoring.guru/design-patterns/builder
    - https://refactoring.guru/design-patterns/proxy
    - https://refactoring.guru/design-patterns/chain-of-responsibility
    - https://refactoring.guru/design-patterns/mediator
2. **CIS 476 Lectures / Lecture Notes**
    - https://canvas.umd.umich.edu/courses/547926/files/folder/Lecture%20Notes

5.2 Technologies & Frameworks

1. **Python Documentation** - https://docs.python.org/3/
    - hashlib (SHA-256 hashing): https://docs.python.org/3/library/hashlib.html
    - uuid: https://docs.python.org/3/library/uuid.html

2. **FastAPI Documentation** - https://fastapi.tiangolo.com/
    - API routing and request handling

3. **SQLAlchemy Documentation** - https://docs.sqlalchemy.org/
    - ORM for SQLite database operations

4. **React Documentation** - https://react.dev/
    - Component-based UI development
    - Context API for state management
    - Hooks (useState, useEffect, useCallback)

5. **React Router** - https://reactrouter.com/
    - Client-side routing and navigation